

CMPE 346 - Natural Language Processing

Final Project

Submission Deadline: 05.06.2026 12.00

Presentation: 05.06.2026 16.00

RULES

- This is a group study. You must work in groups containing 2 or 3 people (no exception).
- Late submissions will **not** be accepted. The submission system will be closed after the deadline. It is highly recommended that you submit your progress at least one day before the deadline. You can change your submissions freely up to the deadline.
- Select a single person to submit on behalf of the group. Multiple submissions in a group would cause 100% similarity in the evaluation metrics. In such cases, your project will not be evaluated.

Submission

- There are three slots for term project submission: **Report Submission**, **Code Submission** and **Presentation Submission**. Submit your files to the related slots. Note that **only** the students who completed these three submissions and made their presentation at the end of the semester will be evaluated.
- Submit your report as [group_id_cmpe346_final].pdf. **Only** PDF file produced from the LaTeX template is accepted for reports. Another format or file type will **not** be accepted.
- Submit your codes as [group_id_cmpe346_final].zip. Do not share your models in your submissions. Submit only the codes that you run for training your models, executing your experiments, etc.
- Submit your presentation slides as [group_id_cmpe346_final_presentation].pdf. **Only** PDF file is accepted for presentations.

Reports

- You are expected to prove the originality of your project. Turnitin will assess the similarity of your report by comparing in-class and online submissions.
- The reports which has more than **30%** similarity (including AI generation score) will **not** be evaluated.

Note that you can see only the similarity score compared to the Internet. In-class and AI similarity will be reported only to the instructor. Since these similarities are evaluated after the deadline, note that your similarity score may increase after the deadline. If you are close to the threshold, it is recommended to revise your report.

- Reports must include at least **1000 words**, excluding titles and references.
- Your report must be well organized and written in clear and precise English. Your grade may be reduced due to obscure or unclear expressions.

Codes

- Your codes must be well organized. The grade for code submission will depend on the organization of your codes.
- Submit your codes in a **zip** file. Do not submit your models and their weights and the datasets.
- Your codes must be clear and clean written. For necessary parts, include documentation. Use classes and separate files. Do not share any Jupyter Notebooks containing uncompleted code and their arbitrary executions.

Presentation

- Last week of the semester (at 05.06.2026), you will make **Presentation** of your project.
- The students who miss the presentations will **NOT** be evaluated.
- Submit your presentations as PDF file. In your slides, do not use more than 2 sentences for each slide. Your presentations must be in **at most** 20 slides.
- Each group will have **5 minutes** to present their projects. Due to the time limitations there will **NOT** be any extra time given.
- Presentations exceeding 5 minutes will be stopped. Be prepared before your presentation. If a student did not present due to the time limitation, this student will **NOT** get any point from presentation.
- During your presentation, do **NOT** have any reading material with you, as well as, do **NOT** read the slides. Students just reading slides or papers will not get any points from presentation.

QUESTION

1. As a group, select one fundamental task (e.g. classification, regression, clustering, generation, sequence labeling, etc.)
2. As an individual, select one utilization of your group task in the NLP domain. Find three different datasets according to your selection.
3. As a group, determine five different deep learning architectures and pre-trained models.
4. Fine-tune the group-selected models in individually selected tasks and report your finding.

EXAMPLE

Assume that **Group 1** contains two students **STU1** and **STU2**.

1. **Group 1** decides their fundamental task as *classification*.
2. **STU1** and **STU2** must find a NLP *classification* task. Assume that **STU1** selects *sentiment analysis*, **STU2** selects *news classification*. Each student must find three different datasets for their task.
3. **Group 1** must select five different DL models: (1) LSTM, (2) BERT, (3) t5, (4) DistilBERT, (5) mBART.
4. Each student must fine-tune the models selected in step-3 by using the datasets selected in step-2.

As a result, each student must fill in a table similar to the one given below.

Model	Dataset 1		Dataset 2		Dataset 3	
	Acc.	F1	Acc.	F1	Acc	F1
LSTM	—	—	—	—	—	—
BERT	—	—	—	—	—	—
t5	—	—	—	—	—	—
DistilBERT	—	—	—	—	—	—
mBART	—	—	—	—	—	—

GRADING

Overall grading of the Term Project is given below. Although it is a group project, note that grading will be individual.

Report:	40%
Codes:	40%
Presentation:	10%
Diversity:	10%

Report

Your report must consist of five sections. The partial gradings regarding the sections are given in the next page.

1. **Introduction:** Comprehensively giving insights about the selected tasks and their related works.
2. **Methodology:** Precise explanation of the models utilized in the project.
3. **Experiments and Findings:** Details about the dataset, the applied configurations and results of the fine-tuned models with evaluation metrics.
4. **Discussion:** Discussion about the results and comparison of models in the selected tasks.
5. **Conclusion:** Summary of the study and thoughts on your findings.

Codes

Your codes must be well organized. Depending on the organization, your codes will be evaluated according to the criteria given below

- (40 points) Codes are well organized and all experiments are completed.
- (30 points) Codes are well organized but all experiments are not completed.
- (20 points) Codes are not clean but all experiments are completed.
- (10 points) Codes are not clean and all experiments are not completed.

Presentation

Each student must participate in the presentation and present their studies. There is a time limit for each group. If a group exceeds the time limit, their presentation will be halted. Presentation criteria is given below

- (10 points) Student presented well in time limits.
- (0 points) Student read slides or papers. / Student did not present due to time limit.

Diversity

This project promotes recent and exceptional studies in NLP domain. Therefore, the selected tasks, which are diverged from other projects and common NLP tasks, will be awarded. For example, subjects utilizing agentic systems, RAG, author anonymization, etc. will get 10 points. In contrast, common tasks such as sentiment analysis will not receive any points from this section of the grading.

Grading Criteria of the Project Report

Section	Incomplete (0pt)	Adequate (5pt)	Good (8pt)
Introduction	Not given or poorly written.	General idea is given about the selected tasks.	The explanation of the field is supported by a few studies conducted in the domain.
Methodology	Not given or poorly written.	The overall methodology is summarized.	The independent units constituting the selected architectures are explained respectively.
Experiments and Findings	Not given or poorly written.	The results of the conducted experiments on the proposed methodology are given.	The experiment configurations are explained in detail along with the results.
Discussion	Not given or poorly written.	The findings of the study are discussed from the perspective of the success of the experiments.	The effects of constituent units in the proposed method are discussed respectively.
Conclusion	Not given or poorly written.	The study is summarized comprehensively.	The project is summarized comprehensively and thoughts about findings are given.